

Published on Education for Global Development

From chalkboards to chatbots: Transforming learning in Nigeria, one prompt at a time

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This page in: English



Students participate in an AI after-school program in Edo, Nigeria. Copyright: SmartEdge/World Bank

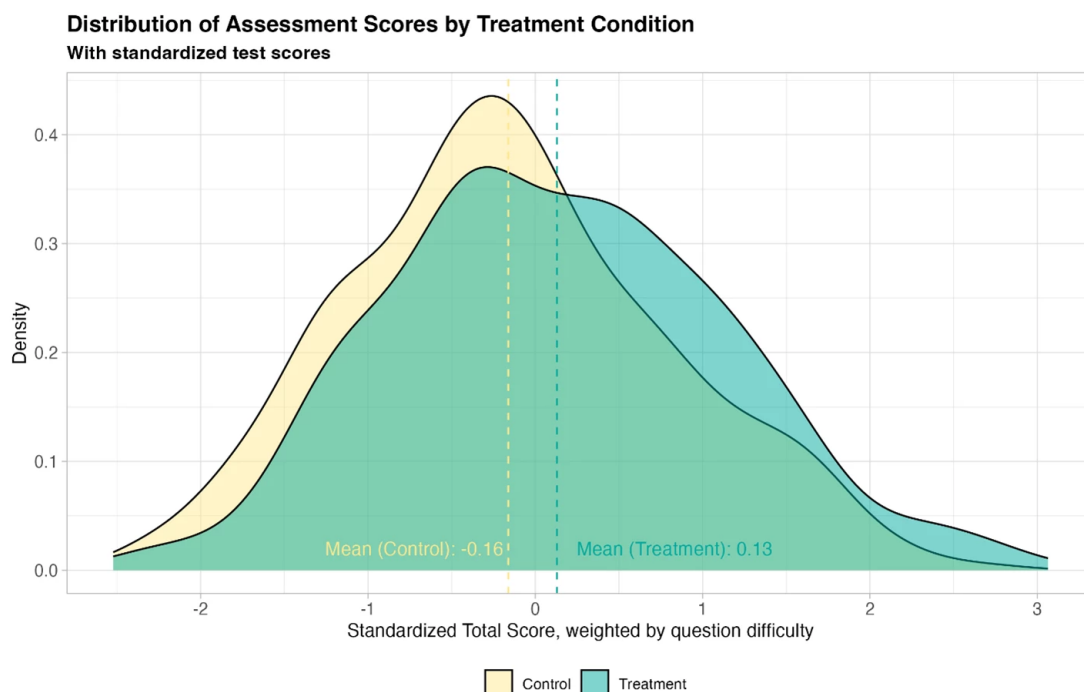
"AI helps us to learn, it can serve as a tutor, it can be anything you want it to be, depending on the prompt you write," says Omorogbe Uyiosa, known as "Uyi" by his friends, a student from the Edo Boys High School, in Benin City, Nigeria. His school was one of the beneficiaries of a pilot that used generative artificial intelligence (AI) to support learning through an after-school program.

A few months ago, we wrote [a blog](#) with some of the lessons from the implementation of this innovative program, including [a video](#) with voices from beneficiaries, such as Uyi. Back then, we promised that, if you stayed tuned, we would get back with the results of the pilot, which included an impact evaluation. So here we are with three primary findings from the pilot!

1. The program boosted learning across the board

The results of the randomized evaluation, soon to be published, reveal overwhelmingly positive effects on learning outcomes. After the six-week intervention between June and July 2024, students took a pen-and-paper test to assess their performance in three key areas: English language—the primary focus of the pilot—AI knowledge, and digital skills.

Students who were randomly assigned to participate in the program significantly outperformed their peers who were not in all areas, including English, which was the main goal of the program. These findings provide strong evidence that generative AI, when implemented thoughtfully with teacher support, can function effectively as a virtual tutor.



Notably, the benefits extended beyond the scope of the program itself. Students who participated also performed better on their end-of-year curricular exams. These exams, part of the regular school program, covered topics well beyond those addressed in the six-week intervention. This suggests that students who learned to engage effectively with AI may have leveraged these skills to explore and master other topics independently.

Moreover, the program benefited all students, not just the highest achievers. Girls, who were initially lagging boys in performance, seemed to gain even more from the intervention, highlighting its potential to bridge gender gaps in learning.

2. Deeper engagement delivered bigger gains

The more sessions students attended, the greater their gains. As discussed in the first blog, attendance was challenging for many students due to factors like flooding during the rainy season, teacher strikes, and after-school work commitments. Using a robust monitoring system developed for the program, we tracked attendance carefully. Each additional day of attendance resulted in significant improvements in learning outcomes. Importantly, the benefits did not taper off as the program progressed. This suggests that a longer program might lead to even greater gains

3. Learning gains were striking

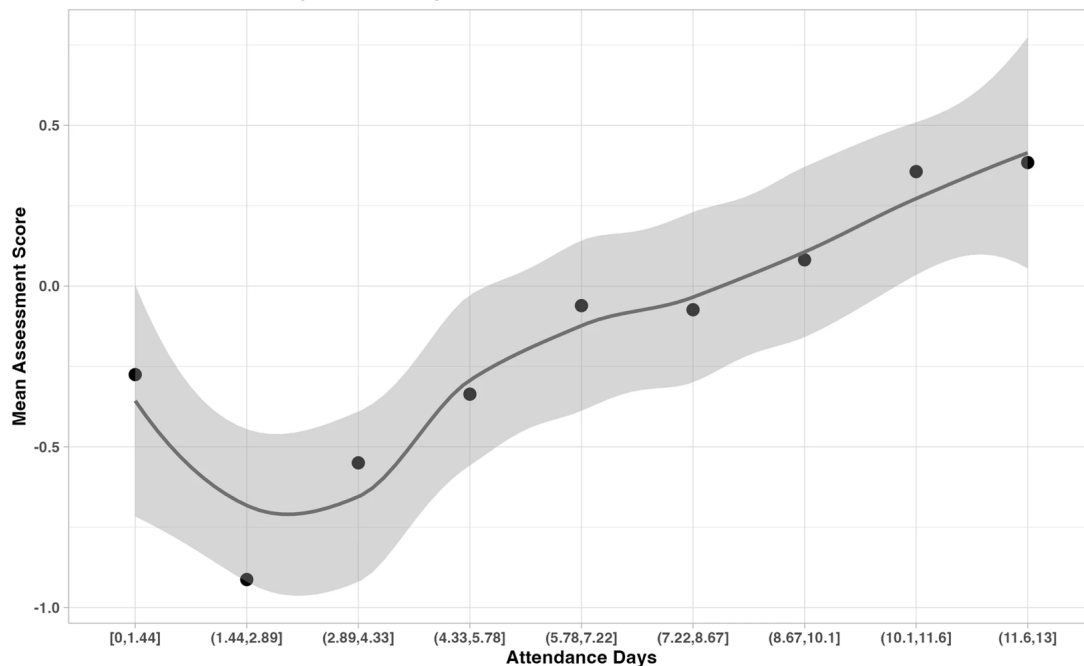
The learning improvements were striking—about 0.3 standard deviations. To put this into perspective, this is [equivalent](#) to nearly two years of typical learning in just six weeks. When we compared these results to a [database of education interventions](#) studied through randomized controlled trials in the developing world, our program outperformed 80% of them, including some of the most cost-effective strategies like structured pedagogy and teaching at the right level. This achievement is particularly remarkable given the short duration of the program and the likelihood that our evaluation design underestimated the true impact.

What's next?

Our evaluation demonstrates the transformative potential of generative AI in classrooms, especially in developing contexts. ✕ To our knowledge, this is the first study to assess the impact of generative AI as a virtual tutor in such settings, building on promising evidence from other contexts and formats; for example, on [AI in coding classes](#), [AI and learning](#) in one school in Turkey, teaching [math with AI](#) (an example through WhatsApp in [Ghana](#)), and [AI as a homework tutor](#).

Dose-Response Relation

Mean assessment score by number of days of attendance



However, this is just the beginning. Several critical questions remain: What are the long-term effects of this intervention? How are students benefiting beyond immediate learning gains? How do their interactions with large language models (LLMs) evolve, and what role do teachers play in supporting these interactions? Are the benefits extending to other subjects? Are there any negative, undesired effects?

Addressing these questions is essential for scaling up similar programs effectively. Stay tuned for the third and final blog in this series, where we'll explore why the key is to use generative AI as a tutor that empowers learning, rather than a shortcut that circumvents it. In the meantime, you can watch this [recent discussion](#) about the potential of AI in education that we had during the recent World Bank Annual Meetings.

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